

Decision and Risk Analysis

Lecture 2: **Bayes' Rule**

DAEN 427/ISEN 427/ISEN 627

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Fake News Detection on Social Media: A Data Mining Perspective

[Kai Shu](#), [Amy Silva](#), [Suhang Wang](#), [Jiliang Tang](#), [Huan Liu](#)

Social media for news consumption is a double-edged sword. On the one hand, its low cost, easy access, and rapid dissemination of information lead people to seek out and consume news from social media. On the other hand, it enables the wide spread of "fake news", i.e., low quality news with intentionally false information. The extensive spread of fake news has the potential for extremely negative impacts on individuals and society. Therefore, fake news detection on social media has recently become an emerging research that is attracting tremendous attention. Fake news detection on social media presents unique characteristics and challenges that make existing detection algorithms from traditional news media ineffective or not applicable. First, fake news is intentionally written to mislead readers to believe false information, which makes it difficult and nontrivial to detect based on news content; therefore, we need to include auxiliary information, such as user social engagements on social media, to help make a determination. Second, exploiting this auxiliary information is challenging in and of itself as users' social engagements with fake news produce data that is big, incomplete, unstructured, and noisy. Because the issue of fake news detection on social media is both challenging and relevant, we conducted this survey to further facilitate research on the problem. In this survey, we present a comprehensive review of detecting fake news on social media, including fake news characterizations on psychology and social theories, existing algorithms from a data mining perspective, evaluation metrics and representative datasets. We also discuss related research areas, open problems, and future research directions for fake news detection on social media.

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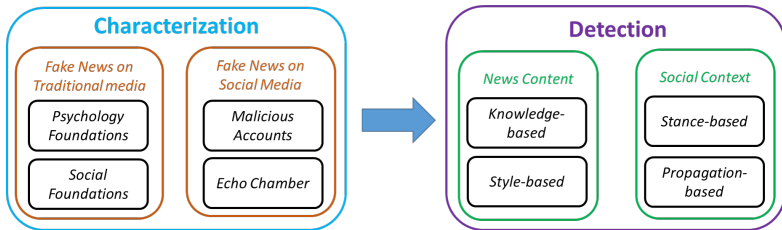
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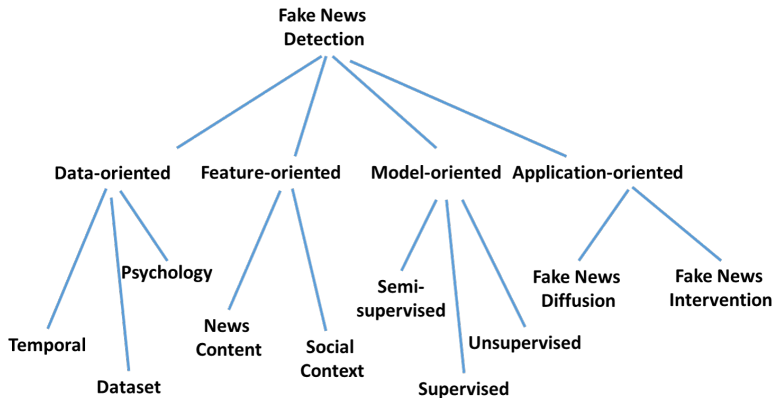
<https://arxiv.org/abs/1708.01967> for details.

Framework



<https://arxiv.org/abs/1708.01967> for details.

Future Directions



<https://arxiv.org/abs/1708.01967> for details.

Building a Bayesian Model for Events

Before reading anything,
we can use prior knowledge from historical data.

Valid Probability Model

A probability model must satisfy:

Building a Bayesian Model for Events

Before reading anything,
we can use prior knowledge from historical data.

Valid Probability Model

A probability model must satisfy:

1. Cover all possible events.
2. Assign probabilities to each event.
3. Probabilities sum to one.

We're Using R Instead of Python

Python

- "Can do everything" ... installs n packages, breaks after update
- Stack Overflow becomes your TA

R → <https://cran.r-project.org>

- Literally for statistics; `lm(y ~ x)` and move on with your life
- Excellent, open-source, keeps us focused on data analysis, ... and I enjoy sleeping at night!

What I Need From You: learn concepts, analyze data, pass the class!

What I Do Not Need: a 2-hour debate about virtual environments.

Dataset

```
1  # Load packages
2  library(bayesrules)
3  library(tidyverse)
4  library(janitor)
5
6  # Import article data
7  data(fake_news)
8
9  fake_news %>%
10     tabyl(type) %>%
11     adorn_totals("row")
12
13  #   type    n percent
14  #   fake   60     0.4
15  #   real   90     0.6
16  # Total 150     1.0
```

Prior Probability Model

- 150 news articles:
60 fake, 90 real
- $P(\text{Fake}) = \frac{60}{150} = 0.40$.
Before looking at the headline,
the prior probability that an
article is fake is 40%.

Prior Belief: there is a 40% chance an article is fake before observing any evidence.

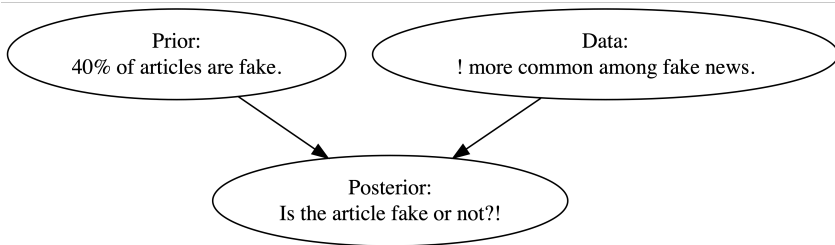
$$P(B) = 0.40 \quad \leftrightarrow \quad P(B^c) = 0.60$$

where B : article is **fake**, B^c : article is **real**.

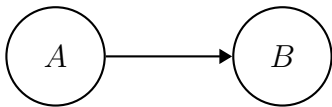
$$P(B) + P(B^c) = 0.40 + 0.60 = 1$$

Event	B	B^c	Total
Probability	0.4	0.6	1

Bayesian reasoning to estimate whether a news article is fake:



DAG: Directed Acyclic Graph



A = exclamation points "!" B = fake status "🤖"

- A **DAG** shows how variables are connected.
- The arrow means A may be associated with B .
- **Acyclic** means the arrows do not loop back.

DAGs and Conditional Probability

- The DAG suggests that the chance of B may depend on A .

$$P(B \mid A)$$

In words: the probability of article is fake **given**
the exclamation point in the title.

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then the events are **dependent**.

$$P(B \mid A) \neq P(B)$$

DAGs and Conditional Probability

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$$P(B \mid A)$$

In words: the probability of article is fake **given**
the exclamation point in the title.

- If A changes the probability of B ,
then the events are **dependent**.

$$P(B \mid A) \neq P(B)$$

- If knowing A does not change the probability of B ,
then they are **independent**.

$$P(B \mid A) = P(B)$$

```
1 # Transposed version of head()
2 # ... makes possible to see all columns in a data frame
3 glimpse(fake_news)
```

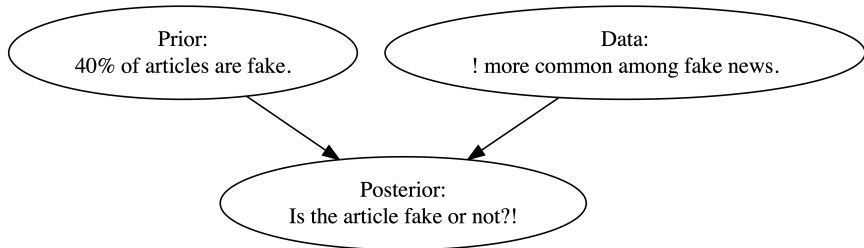
Observed headline pattern:

```
1 # Tabulate exclamation usage and article type
2 fake_news %>%
3   tabyl(title_has_excl, type) %>%
4   adorn_totals("row")
5 # title_has_excl fake real
6 #           FALSE  44   88
7 #           TRUE   16    2
8 #           Total  60   90
```



Code: <https://abuabara.github.io/DAEN427F26/Lecture2.R>

Exclamation Points as Evidence



Title Has "!"	Fake	Real	Total
Yes	16	2	18
No	44	88	132
Total	60	90	150

Conditional Probability & Likelihood

Title Has "!"	Fake	Real	Total
Yes	16	2	18
No	44	88	132
Total	60	90	150

$$P(! \mid \text{Fake}) = \frac{16}{60} \approx 0.267, P(! \mid \text{Real}) = \frac{2}{90} \approx 0.022$$

An exclamation point increases the probability that an article is fake.

Conditional Probability & Likelihood

B : article is fake

B^c : article is real

A : article uses exclamation points

A^c : article does not use exclamation points

From the data:

$$P(A \mid B) = 0.267 \quad P(A \mid B^c) = 0.022$$

Fake news articles are **much more likely** to use exclamation points than real news articles.

Conditional vs Unconditional Probability

- **Unconditional probability** is the chance something happens without extra information.

$$P(A)$$

- **Conditional probability** is the chance something happens given that something else is already known.

$$P(A \mid B)$$

- Comparing $P(A)$ and $P(A \mid B)$ tells us whether knowing B changes how likely A is.

Independent Events

- Two events are **independent** if knowing one happens does not change the probability of the other.
- *Example*: rolling a die and flipping a coin are independent, because the die result **does not** affect the coin flip.
- In symbols:

$$P(A \mid B) = P(A)$$

From Probability to Likelihood

- Earlier, we computed:

$$P(A \mid B) = 0.2667 \quad P(A \mid B^c) = 0.0222$$

- Fake articles are much more likely to use exclamation points.
- Suppose we already observed an article with exclamations.

Now the question becomes:

Is the article more likely fake or real?

What is Likelihood?

- **Probability** starts with a known cause and predicts data.

$$P(A \mid B)$$

- **Likelihood** starts with observed data and compares possible causes.

$$\mathcal{L}(B \mid A) = P(A \mid B)$$

- In our example:
 - A : article uses exclamation points
 - B : article is fake
 - Since

$$P(A \mid B) > P(A \mid B^c)$$

exclamation provide evidence that the article may be fake.

Probability	Likelihood
Known condition	Known data
Predicts outcomes	Compares explanations
$P(A \mid B)$	$\mathcal{L}(B \mid A)$

- The distinction is **subtle**, especially since people use the terms “likelihood” and “probability” interchangeably in casual conversation.
- Likelihood **helps us decide** which explanation best fits the observed data.

- The likelihood function **is not** a probability function, but rather provides a framework to compare the relative compatibility of our exclamation point data with B and B^c .

Event	B	B^c	Total
Prior probability	0.4	0.6	1
Likelihood ($\cdot \mid A$)	0.2667	0.0232	0.2889

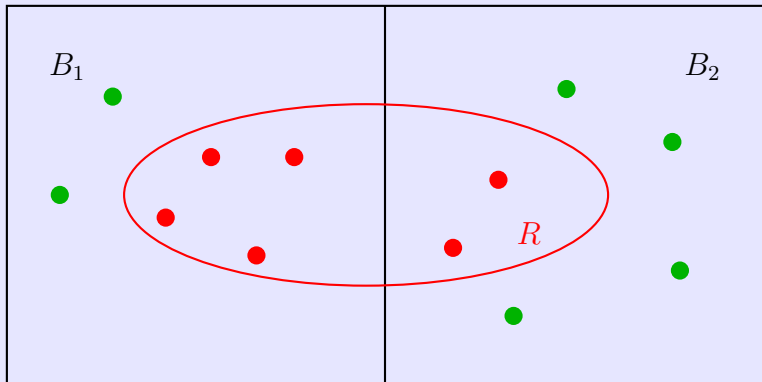
Quiz

Two boxes are available:

- Box 1 contains 4 red and 2 green balls.
- Box 2 contains 2 red and 4 green balls.
- The probability of choosing either box is equal:

$$P(E_1) = P(E_2) = \frac{1}{2}.$$

- A box is selected at random, and one ball is drawn.
- Study the **likelihood** of observing a red ball under each box.



Answer: A red ball is more consistent with Box 1 than with Box 2.

The likelihoods are: $P(R \mid B_1) = \frac{4}{6} = \frac{2}{3}$ and $P(R \mid B_2) = \frac{2}{6} = \frac{1}{3}$.

So, if Box 1 had been selected, observing a red ball would be more likely.

$$P(R \mid B_1) > P(R \mid B_2)$$

Therefore, the red ball provides stronger evidence for Box 1 than for Box 2.

$$\frac{P(R \mid B_1)}{P(R \mid B_2)} = \frac{\frac{2}{3}}{\frac{1}{3}} = 2$$

The observed red ball is **twice** as likely under Box 1 as under Box 2. Since the boxes were equally likely before the draw, the red ball favors Box 1.

Recap

A = article uses exclamation points

A^c = article does not use exclamation points

B = article is fake news

B^c = article is real news

From the prior model:

$$P(B) = 0.4 \quad P(B^c) = 0.6$$

So, 40% of articles are fake and 60% are real.



Normalizing Constants

- The marginal probability of observing exclamation points across all articles is $P(A)$.
- This is called the **normalizing constant**.
- To calculate it, we split articles into two groups:

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

- That is:
Exclamations = fake with exclamations + real with exclamations

Joint Probability Table

	B	B^c	Total
A	$P(A \cap B)$	$P(A \cap B^c)$	$P(A)$
A^c	$P(A^c \cap B)$	$P(A^c \cap B^c)$	$P(A^c)$
Total	0.4	0.6	1

- Each inside cell is a **joint probability**.
- For example: $P(A \cap B)$ means the probability that an article is fake and uses exclamation points.

Fake Articles **with** Exclamation Points

- To determine the probabilities of these joint events, note that 40% of articles are fake and 26.67% of fake articles use exclamation points, $P(B) = 0.4$ and $P(A | B) = 0.2667$.
- The joint probability is
$$P(A \cap B) = P(A | B) P(B) = 0.2667 \cdot 0.4 = 0.1067$$
- So, 10.67% of all articles are fake and use exclamation points.

	B	B^c	Total
A	$0.4 \cdot 16/60 = \mathbf{0.107}$	$P(A \cap B^c)$	$32/150 = 0.12$
A^c	$P(A^c \cap B)$	$P(A^c \cap B^c)$	$118/150 = 0.78$
Total	0.4	0.6	1

Fake Articles **without** Exclamation Points

- Since 26.67% of fake articles use exclamation points,
 $1 - 0.2667 = 73.33\%$ do not use exclamation points.
- Then,
$$P(A^c \cap B) = P(A^c \mid B) P(B) = 0.7333 \cdot 0.4 = 0.2933$$
- So, 29.33% of all articles are fake and do not use exclamation.

Checking the Fake Article Total

The fake-news column is split into two parts:

$$P(A \cap B) \quad \text{and} \quad P(A^c \cap B)$$

Therefore,

$$P(B) = P(A \cap B) + P(A^c \cap B)$$

$$P(B) = 0.1067 + 0.2933$$

$$P(B) = 0.4$$

This matches the prior probability that an article is fake.

Real Articles

From the likelihood function: $P(A \mid B^c) = 0.0222$

So only 2.22% of real articles use exclamation points.

The joint probability is

$$P(A \cap B^c) = P(A \mid B^c)P(B^c)$$

$$P(A \cap B^c) = 0.0222 \cdot 0.6$$

$$P(A \cap B^c) = 0.0133$$

So, 1.33% of all articles are real and use exclamation points.

Real Articles without Exclamation Points

Since

$$P(A \mid B^c) = 0.0222$$

we have

$$P(A^c \mid B^c) = 1 - 0.0222$$

$$P(A^c \mid B^c) = 0.9778$$

Then

$$P(A^c \cap B^c) = P(A^c \mid B^c) P(B^c)$$

$$P(A^c \cap B^c) = 0.9778 \cdot 0.6$$

$$P(A^c \cap B^c) = 0.5867$$

Completed Joint Probability Table

	B	B^c	Total
A	0.107	0.013	0.12
A^c	0.293	0.587	0.88
Total	0.4	0.6	1

The row total for A gives the **normalizing constant**:

$$P(A) = 0.107 + 0.013 = \boxed{0.12}$$

So, 12% of all articles use exclamation points.

Joint Probability

The calculations use the relationship

$$P(A \cap B) = P(A \mid B)P(B)$$

This connects: joint probability

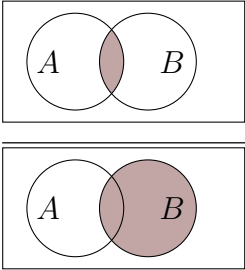
with: conditional probability $\times$ prior probability

Rearranging gives the conditional probability formula:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)}$$

PS. If A and B are **independent**, then: $P(A \cap B) = P(A) P(B)$

Representation of Conditional Probability

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} = \frac{\text{Diagram 1}}{\text{Diagram 2}}$$


The image shows two Venn diagrams within rectangular frames. The top diagram contains two overlapping circles, A and B. The intersection of A and B is shaded in a light brown color. The bottom diagram also contains two overlapping circles, A and B. In this diagram, the entire circle B is shaded in the same light brown color, including its intersection with circle A.

Conditional Probability

Conditional probability answers the question:

"What is the probability of A given that B has occurred?"

The definition is:

$$P(A \mid B) = \frac{P(A \cap B)}{P(B)} \quad \text{where } P(B) \neq 0$$

- Start with the probability that both A and B occur
- Restrict attention only to situations where B occurs
- Measure how often A also occurs

Example: $P(\text{Rain} \mid \text{Cloudy}) = \frac{P(\text{Rain and Cloudy})}{P(\text{Cloudy})}$

Law of Total Probability

Suppose event A can happen in two distinct ways:

$$A \cap B \quad \text{or} \quad A \cap B^c$$

Since B and B^c cover all possibilities, we can write:

$$P(A) = P(A \cap B) + P(A \cap B^c)$$

Using the joint probability rule:

$$P(A \cap B) = P(A \mid B)P(B)$$

we get the **Law of Total Probability**:

$$P(A) = P(A \mid B)P(B) + P(A \mid B^c)P(B^c)$$

Exclamation Points in News Articles

A = article uses an exclamation point

B = article is fake news

$$P(A | B) = 0.2667, \quad P(B) = 0.4$$

$$P(A | B^c) = 0.0222, \quad P(B^c) = 0.6$$

Then:

$$P(A) = 0.2667 (0.4) + 0.0222 (0.6)$$

$$P(A) = 0.1067 + 0.0133 = 0.12$$

So, about **12% of all articles** use exclamation points.

	B	B^c	Total
A	0.107	0.013	0.12
A^c	0.293	0.587	0.88
Total	0.4	0.6	1

$$P(A) = 0.107 + 0.013 = \boxed{0.12}$$

Homework

- Questions 1–5 next; questions 6-9 at the end of the topic.
- Submit all answers by Fri Sep 11.
- You may discuss with peers but submissions must be individual.
- Responses may be shared anonymously during class discussion!

Question 1

For each scenario below, you're given a pair of events, A and B . Explain what you believe to be the relationship between the posterior and prior probabilities of B : $P(B|A) > P(B)$ or $P(B|A) < P(B)$.

- a) A = you just finished reading Lambda Literary Award-winning author Nicole Dennis-Benn's first novel, and you enjoyed it!
 B = you will also enjoy Benn's newest novel.
- b) A = it's 0 degrees Fahrenheit in Minnesota on a January day.
 B = it will be 60 degrees tomorrow.
- c) A = the authors only got 3 hours of sleep last night.
 B = the authors make several typos in their writing today.
- d) A = your friend includes three hashtags in their tweet.
 B = the tweet gets retweeted.

Question 2

The probability that Gabby will like a restaurant is 0.7. Among the restaurants that she likes, 20% have five stars on Yelp, 50% have four stars, and 30% have fewer than four stars. What other information do we need if we want to find the posterior probability that Gabby likes a restaurant given that it has fewer than four stars on Yelp?

Question 3

Matt is on a dating app looking for love. Matt swipes right on 8% of the profiles he views. Of the people that Matt swipes right on, 40% are men, 30% are women, 20% are non-binary, and 10% identify in another way. Of the people that Matt does not swipe right on, 45% are men, 40% are women, 10% are non-binary, and 5% identify in some other way.

- a) What's the probability that a randomly chosen person on this dating app is non-binary?
- b) Given that Matt is looking at the profile of someone who is non-binary, what's the posterior probability that he swipes right?

Question 4

For a certain airline, 30% of the flights depart in the morning, 30% depart in the afternoon, and 40% depart in the evening.

Frustratingly, 15% of all flights are delayed.

Of the delayed flights, 40% are morning flights, 50% are afternoon flights, and 10% are evening flights.

Alicia and Mine are taking separate flights to attend a conference.

- a) Mine is on a morning flight. What's the probability that her flight will be delayed?
- b) Alicia's flight is not delayed. What's the probability that she's on a morning flight?

Question 5 (answer revealed on the next slide!)

A medical test correctly detects a disease 95% of the time when the disease is present. This value represents:

- a. Prior probability
- b. Posterior probability
- c. False positive rate
- d. Likelihood

Answer: d. Likelihood

The statement describes: $P(\text{Test positive} \mid \text{Disease present}) = 0.95$

This is the probability of observing the test result given that the disease is present, which represents the likelihood (also called **sensitivity** or **true positive rate**).

- **Prior:** Probability of disease before observing the test result.
- **Posterior:** Probability of disease after observing the test result.
- **False positive rate:** Probability that the test is positive when the disease is absent.



From Events to Random Variables

The previous example asked about a categorical outcome: fake or real?

In 1996, world chess champion Garry Kasparov defeated IBM Deep Blue in a six-game match, winning three games, drawing two, and losing one.

The rivals **met again** in 1997 for another six-game showdown.

- Let π denote the chances of a player (Kasparov) winning any particular game in this six-game re-match.
- Thus, π is a measure of his overall skill relative to the machine.
- Given the complexity of chess, machines, and humans, π is unknown and can vary or fluctuate over time.
- Or, in short, π is a random variable.
Now the outcome of interest is numerical!

Building a Bayesian Model for Random Variables

- **Prior:** What we believe about π before seeing the data.
- **Data model:** How likely different data values are for a given value of π .
 - We'll collect and process data which can inform our understanding of π (Kasparov's skill level relative to that of Deep Blue).
 - Our data Y is the number of the six games in the 1997 re-match that Kasparov wins.
 - Y is a random variable that can take any value in $\{0, 1, 2, 3, 4, 5, 6\}$.
- **Posterior:** What we believe about π after seeing the data.

Building a Bayesian Model for Random Variables



Bayesian analysis is a formal way of learning from data.

π represents the probability of winning a chess game. We begin with an initial belief about π , observe some games, and then update our belief based on the results.

Kasparov vs. Deep Blue

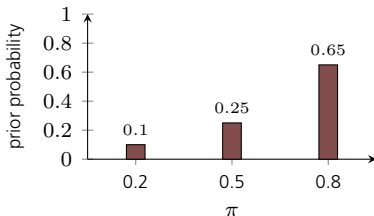
In a six-game re-match, we want to learn about:

π = Kasparov's probability of winning one game

The prior probability π can technically be any number from 0 to 1.

For teaching purposes, let us assume a discrete set of possibilities:

π	0.2	0.5	0.8	Total
$f(\pi)$	0.10	0.25	0.65	1.00



Discrete Probability Model

Y is a discrete random variable.

The probability model of Y is specified by a **probability mass function (pmf)** $f(y)$.

This pmf defines the probability of any given outcome y ,

$$f(y) = P(Y = y)$$

and has the following properties:

- $0 \leq f(y) \leq 1$ for all y ; and
- $\sum_{\text{all } y} f(y) = 1$, i.e., probabilities of all possible outcomes y sum to 1.

Here, $f(\pi)$ is the **probability mass function** of the prior distribution for π .

$$f(\pi) = P(\Pi = \pi)$$

Thus,

$$P(\Pi = 0.2) = 0.10, \quad P(\Pi = 0.5) = 0.25, \quad P(\Pi = 0.8) = 0.65.$$

Because this is a **discrete prior**,

$f(\pi)$ tells us how much prior probability we assign to each possible value of π .

What is a Probability Mass Function?

For a discrete random variable, a **probability mass function (pmf)** gives probability to each possible value.

PMF Requirements

For possible values y ,

$$f(y) = P(Y = y), \quad 0 \leq f(y) \leq 1, \quad \sum_{\text{all } y} f(y) = 1.$$

For the chess example, the prior pmf $f(\pi)$ says:

- $f(\pi = 0.8) = 0.65$: before the re-match, high confidence in Kasparov.
- $f(\pi = 0.2) = 0.10$: some chance Deep Blue is much stronger.

Is this a probability of a probability?

Yes, informally:

- π is itself a **probability parameter**:
 $\pi = P(\text{Kasparov wins one game})$
- The prior distribution assigns probabilities to possible values of π .
- For example, $P(\pi = 0.8) = 0.65$ means that:
before observing the new data, we assign a 65% prior probability to the possibility that Kasparov's game-win probability is 0.8.

Prior distribution = probability distribution over an unknown probability parameter

Number of Wins (The Data!)

Let

Y = number of games Kasparov wins in the six-game re-match.

Possible values:

$$Y \in \{0, 1, 2, 3, 4, 5, 6\}.$$

The data model should answer:

If π were 0.2, 0.5, or 0.8, how likely would each possible Y be?

Number of Wins (The Data!)

Let

Y = number of games Kasparov wins in the six-game re-match.

Possible values:

$$Y \in \{0, 1, 2, 3, 4, 5, 6\}.$$

Observed result:

In the 1997 re-match,
Kasparov won 1 of 6 games!

The data model should answer:

*If π were 0.2, 0.5, or 0.8, how
likely would each possible Y be?*

A Binomial model is reasonable when we have:

1. A fixed number of trials: $n = 6$ games.
2. A binary outcome on each trial: win or not win.
3. The same success probability π for each game.
4. Independent game outcomes, conditional on π .

A Binomial model is reasonable when we have:

1. A fixed number of trials: $n = 6$ games.
2. A binary outcome on each trial: win or not win.
3. The same success probability π for each game.
4. Independent game outcomes, conditional on π .

Model Notation

$$Y \mid \pi \sim \text{Bin}(6, \pi)$$

The “vertical bar” means **model the data after assuming a particular value of π** , and $\sim$ can be read as **“modeled by”**.

The Binomial Data Model

For $Y \mid \pi \sim \text{Bin}(n, \pi)$,

$$f(y \mid \pi) = \binom{n}{y} \pi^y (1 - \pi)^{n-y}, \quad y = 0, 1, \dots, n.$$

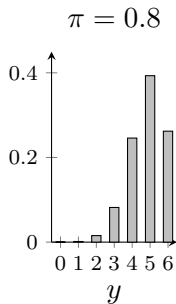
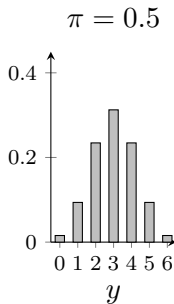
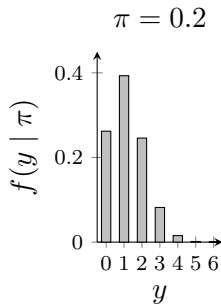
$$\binom{n}{y} = \frac{n!}{y!(n-y)!}$$

$f(y \mid \pi)$ is the conditional probability of observing y wins if the win probability really were π .

For six games:

$$f(y \mid \pi) = \binom{6}{y} \pi^y (1 - \pi)^{6-y}, \quad y = 0, 1, \dots, 6.$$

How the Data Model Changes with π



Observed Data Changes the Question

Before seeing data, we compared possible values of Y under one fixed π :

$$f(y_1 \mid \pi) \quad \text{vs.} \quad f(y_2 \mid \pi).$$

After observing $Y = 1$, the question flips:

$$L(\pi \mid y = 1) = f(y = 1 \mid \pi).$$

The likelihood is a function of π after the data are known.

It tells us how compatible the observed data are with each possible π .

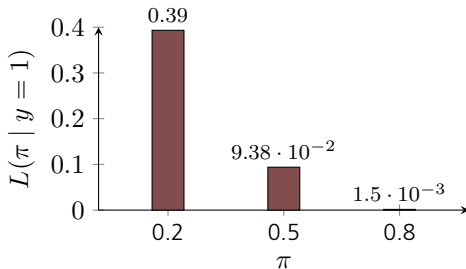
Example: What is the probability of it being cloudy, given that we know it is raining?

Likelihood for 1 Win

For $y = 1$ and $n = 6$:

$$L(\pi \mid y = 1) = \binom{6}{1} \pi (1 - \pi)^5 = 6\pi(1 - \pi)^5.$$

π	0.2	0.5	0.8
$L(\pi \mid y = 1)$	0.3932	0.0938	0.0015



PMF vs. Likelihood: Same Formula, Different Role

The same numerical calculation can play 2 different conceptual roles:

Conditional PMF

π is fixed, Y varies.

$$f(y | \pi) = P(Y = y | \pi)$$

It is a probability model over possible data values, so it sums to 1 over y .

Likelihood function

y is fixed, π varies.

$$L(\pi | y) = f(y | \pi)$$

It compares parameter values. It does **not** have to sum to 1 over π .

Normalizing Constant

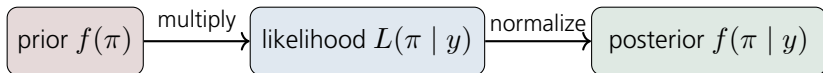
Bayes' rule needs the overall probability of the observed data:

$$f(y = 1) = \sum_{\pi \in \{0.2, 0.5, 0.8\}} f(\pi) L(\pi \mid y = 1).$$

$$f(y = 1) \approx 0.10(0.3932) + 0.25(0.0938) + 0.65(0.0015) \approx 0.0637$$

Why normalize? The products $f(\pi)L(\pi \mid y)$ are evidence scores. Dividing by their total turns those scores into posterior probabilities that sum to 1.

Posterior PMF



$$f(\pi | y) = \frac{f(\pi)L(\pi | y)}{f(y)}$$

For the chess example:

$$f(\pi | y = 1) = \frac{f(\pi)L(\pi | y = 1)}{0.0637}.$$

Practical version

$$\text{posterior probability} = \frac{\text{prior-likelihood product}}{\text{sum of all prior-likelihood products}}$$

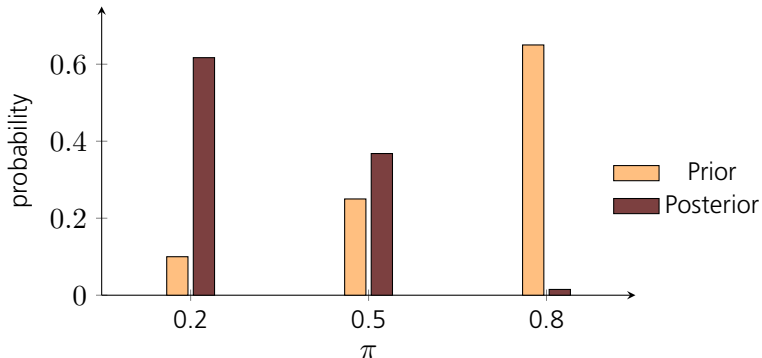
1. Multiply prior by likelihood for each possible π
2. Add the products
3. Divide each product by the total

Posterior Calculation Table

π	0.2	0.5	0.8
Prior $f(\pi)$	0.100	0.250	0.650
Likelihood $L(\pi \mid y = 1)$	0.3932	0.0938	0.0015
Unnormalized $f(\pi)L(\pi \mid y = 1)$	0.0393	0.0234	0.0010
Posterior $f(\pi \mid y = 1)$	0.617	0.368	0.015

After observing only 1 win in 6 games, the posterior places most of its mass on $\pi = 0.2$. The data strongly moved belief away from $\pi = 0.8$.

Prior vs. Posterior



A small amount of surprising data can cause a large update when the likelihood is sharply different across parameter values.

Posterior Shortcut: Proportionality

Because $f(y)$ is the same denominator for every π ,

$$f(\pi | y) = \frac{f(\pi)L(\pi | y)}{f(y)} \propto f(\pi)L(\pi | y).$$

The symbol $\propto$ means “is proportional to.”

In Bayesian statistics, the posterior is found by combining:

$$\text{prior} \times \text{likelihood}.$$

The shortcut is that we do not always need to compute the normalizing constant:

$$\text{posterior} \propto \text{prior} \times \text{likelihood}$$

A probability distribution must add up to 1.

For example, if the possible values of π are

$$\pi \in \{0.2, 0.5, 0.8\},$$

then the posterior probabilities must satisfy

$$f(\pi = 0.2 \mid y = 1) + f(\pi = 0.5 \mid y = 1) + f(\pi = 0.8 \mid y = 1) = 1.$$

The normalizing constant rescales the posterior probabilities so they sum to 1.

Step 1: Compute Unnormalized Posterior Values

Instead of calculating the normalizing constant immediately, multiply the prior by the likelihood.

$$f(\pi = 0.2 \mid y = 1) \propto 0.10(0.3932) = 0.039320,$$

$$f(\pi = 0.5 \mid y = 1) \propto 0.25(0.0938) = 0.023450,$$

$$f(\pi = 0.8 \mid y = 1) \propto 0.65(0.0015) = 0.000975.$$

These values are not probabilities yet because they do not sum to 1.

Step 2: Add the Unnormalized Values

Now add the unnormalized posterior values:

$$0.039320 + 0.023450 + 0.000975 = 0.063745.$$

This total tells us how much we need to divide by to turn the values into real probabilities.

$\text{normalized probability} = \frac{\text{unnormalized value}}{\text{sum of unnormalized values}}$

Step 3: Normalize Each Value

For example, the posterior probability for $\pi = 0.2$ is

$$f(\pi = 0.2 \mid y = 1) = \frac{0.039320}{0.039320 + 0.023450 + 0.000975} \approx 0.617.$$

Similarly,

$$\begin{aligned} f(\pi = 0.5 \mid y = 1) &\approx 0.368, \\ f(\pi = 0.8 \mid y = 1) &\approx 0.015. \end{aligned}$$

Now the posterior probabilities add up to 1.

The General Formula

Bayes' rule can be written as

$$f(\pi \mid y) = \frac{f(\pi)L(\pi \mid y)}{f(y)}.$$

For a discrete set of possible π values, the denominator is

$$f(y) = \sum_{\text{all } \pi} f(\pi)L(\pi \mid y).$$

So we can write

$$f(\pi \mid y) = \frac{f(\pi)L(\pi \mid y)}{\sum_{\text{all } \pi} f(\pi)L(\pi \mid y)}$$

The Posterior Shortcut

Since $f(y)$ does not depend on π , it is just a constant for normalization.

Therefore,

$$f(\pi | y) = \frac{f(\pi)L(\pi | y)}{f(y)} \propto f(\pi)L(\pi | y).$$

In words:

$\text{posterior} \propto \text{prior} \times \text{likelihood}$
--

The posterior shortcut lets us build a Bayesian posterior in two simple steps:

1. Multiply each prior probability by its likelihood.
2. Divide each result by the total so the probabilities sum to 1.

This preserves the relative sizes of the posterior probabilities while turning them into a valid probability distribution.

All the information comes from the prior and the likelihood.

Workflow for Building a Bayesian Model

1. Define the unknown quantity π and its possible values



2. Assign a prior PMF $f(\pi)$



3. Specify the data model $f(y \mid \pi)$



4. Observe data y and compute likelihood $L(\pi \mid y)$



5. Normalize $f(\pi)L(\pi \mid y)$ to obtain $f(\pi \mid y)$

Common Mistakes

- 1)** Treating likelihood values as probabilities over π .
Likelihoods compare compatibility. They do not have to sum to 1.
- 2)** Ignoring the prior once the data arrive.
The posterior balances prior information and data evidence.
- 3)** Forgetting that the Binomial model depends on assumptions:
fixed n , binary outcomes, same π , and conditional independence.

Quiz

Suppose $Y \mid \pi \sim \text{Bin}(6, \pi)$ and the prior is still

$$\pi : 0.2, 0.5, 0.8 \quad f(\pi) : 0.10, 0.25, 0.65.$$

Before calculating anything: if Kasparov had won 5 of 6 games, which value of π would probably get the most posterior support?

Quiz

Suppose $Y \mid \pi \sim \text{Bin}(6, \pi)$ and the prior is still

$$\pi : 0.2, 0.5, 0.8 \quad f(\pi) : 0.10, 0.25, 0.65.$$

Before calculating anything: if Kasparov had won 5 of 6 games, which value of π would probably get the most posterior support?

Answer: $\pi = 0.8$ should get the most support, because $Y = 5$ is much more compatible with a high win probability than a low one.

Simulation Intuition

A simulation version of the same analysis:

1. Draw many π values from the prior model.
2. For each π , simulate $Y \mid \pi \sim \text{Bin}(6, \pi)$.
3. Keep only simulations where $Y = 1$.
4. The distribution of retained π values approximates the posterior.

```
1 pi <- sample(c(.2,.5,.8), 10000, replace=TRUE,  
  ↪ prob=c(.10,.25,.65))  
2 y <- rbinom(10000, size=6, prob=pi)  
3 prop.table(table(pi[y == 5]))
```

Takeaways

1. A Bayesian model can treat an unknown numerical parameter as a random variable.
2. The prior PMF describes what parameter values were plausible before seeing data.
3. The likelihood describes how compatible the observed data are with each parameter value.
4. The posterior PMF updates the prior by multiplying by the likelihood and normalizing.

Maximum Likelihood

- Suppose Kasparov wins only $y = 1$ out of $n = 6$ games.
- Let $\pi = P(\text{Kasparov wins a game})$, with possible values $\pi \in \{0.2, 0.5, 0.8\}$.
- For a binomial model, the likelihood is $L(\pi) \propto \pi^1(1 - \pi)^5$.

π	Relative likelihood $\pi(1 - \pi)^5$
0.2	0.0655
0.5	0.0156
0.8	0.000064

$$\hat{\pi}_{\text{MLE}} = 0.2$$

MLE chooses the value of π that makes the observed data most likely.

Bayesian Updating and Maximum A Posteriori (MAP)

Bayesian inference also considers our **prior belief**:

$$P(\pi \mid y) \propto P(y \mid \pi) P(\pi).$$

In the chess example, the prior probabilities are

$$P(\pi = 0.2) = 0.10, \quad P(\pi = 0.5) = 0.25, \quad P(\pi = 0.8) = 0.65.$$

After observing 1 win in 6 games:

π	0.2	0.5	0.8
$P(\pi \mid y)$	0.617	0.368	0.015

Therefore, $\hat{\pi}_{\text{MAP}} = 0.2$

Key Difference

MLE: maximize likelihood

MAP: maximize likelihood $\times$ prior

In this example, $\hat{\pi}_{\text{MLE}} = \hat{\pi}_{\text{MAP}} = 0.2$
but they do not have to be equal in general.

In-class Mini Exercise

A student applies to 6 internships. Let π be their chance of getting any one offer.

π	0.3	0.4	0.5	Total
$f(\pi)$	0.25	0.60	0.15	1.00

They receive $y = 4$ offers.

1. Write the data model for $Y \mid \pi$.
2. Compute $L(\pi \mid y = 4)$ for each π .
3. Multiply by the prior and normalize.
4. Which π has the highest posterior probability?

1. Data Model

Assuming the internship outcomes are conditionally independent,

$$Y \mid \pi \sim \text{Binomial}(6, \pi).$$

Thus,

$$P(Y = y \mid \pi) = \binom{6}{y} \pi^y (1 - \pi)^{6-y}.$$

For the observed value $y = 4$,

$$L(\pi \mid y = 4) = \binom{6}{4} \pi^4 (1 - \pi)^2.$$

2. Compute the Likelihood in R

Evaluate the likelihood at each possible value of π .

```
pi <- c(0.3, 0.4, 0.5)
```

```
likelihood <- dbinom(  
  x = 4,  
  size = 6,  
  prob = pi  
)
```

```
likelihood
```

π	0.3	0.4	0.5
$L(\pi \mid y = 4)$	0.059535	0.138240	0.234375

Notice that the likelihood alone is largest at $\pi = 0.5$.

3. Multiply by the Prior and Normalize

Bayes' rule for a discrete prior is

$$f(\pi | y) = \frac{L(\pi | y)f(\pi)}{\sum_{\pi} L(\pi | y)f(\pi)}.$$

```
prior <- c(0.3, 0.4, 0.5)
weight <- likelihood * prior
posterior <- weight / sum(weight)
data.frame(pi, prior, likelihood, weight, posterior)
```

π	0.3	0.4	0.5
$L(\pi y)f(\pi)$	0.014884	0.082944	0.035156
$f(\pi y)$	0.112	0.624	0.264

4. Posterior Conclusion

The posterior distribution is

π	0.3	0.4	0.5
$f(\pi \mid y = 4)$	0.112	0.624	0.264

In R, the most probable value is

```
pi[which.max(posterior)]
```

which gives $\pi = 0.4$.

Although $\pi = 0.5$ has the largest likelihood, the prior places much more probability on $\pi = 0.4$.

Therefore,

$$P(\pi = 0.4 \mid Y = 4) \approx 0.624$$

is the largest posterior probability.

Question 6

Muhammad applies for six equally competitive data science internships. His prior model for his probability π of receiving an offer from any given internship is

π	0.3	0.4	0.5
$f(\pi)$	0.25	0.60	0.15

- a) Let Y be the number of internship offers that Muhammad gets. Specify the model for the dependence of Y on π and the corresponding pmf, $f(y|\pi)$.
- a) Muhammad got some pretty amazing news. He was offered four of the six internships! How likely would this be if $\pi = 0.3$?
- c) Construct the posterior model of π in light of Muhammad's internship news.

Question 7

Whether you like it or not, cats have taken over the internet. Joining the craze, Michael has written an algorithm to detect cat images. It correctly identifies 80% of cat images as cats, but falsely identifies 50% of non-cat images as cats. Michael tests her algorithm with a new set of images, 8% of which are cats. What's the probability that an image is actually a cat if the algorithm identifies it as a cat? Answer this question by simulating data for 10,000 images.

Question 8 (no spoilers, next slide has the answer!)

A university uses an AI system to detect plagiarism.

- 5% of assignments are actually plagiarized.
- The AI flags plagiarized assignments 92% of the time.
- The AI incorrectly flags honest assignments 8% of the time.

If an assignment is flagged, what is the probability it was actually plagiarized? Briefly discuss whether the AI system seems reliable.

Answer:

$$P(P \mid F) = \frac{0.92 \cdot 0.05}{0.92 \cdot 0.05 + 0.08 \cdot 0.95} \approx 0.377$$

Approximately 37.7% chance that it is actually plagiarism.

The system catches most plagiarism cases, but because plagiarism is rare, many flagged assignments are false positives.

Question 9



Let's suppose that you are an air traveler along with 99 other passengers from your flight, each of whom, like you, checked one item of luggage.

A recording piped through the speakers reminds you that: *"Many bags look alike. Please check your bag carefully before exiting the terminal."*

Indeed, your bag is one of the most popular models on the market, a medium-sized black rectangular case **used by 5% of all travelers**.

Given your distance from the luggage chute, you can discern only the shape, size, and color of the bags that emerge, not any detailed markings on the bags.

Question 9, cont.

- a. Let's suppose that the **first bag** from your flight to enter the luggage carousel indeed has the same shape, size, and color as yours. Is it yours? Explain (i.e., calculate the posterior probability that the 1st bag is yours).
- b. Now suppose that we continue to wait for our bag to appear, failing to see it among the first 98 bags to enter the carousel (i.e., only 2 bags are left to come off the airplane). Calculate the posterior probability that the **99th** bag, which also matches ours in shape, size, and color, is our own.

(maybe the answers will follow ...)

Answer: a. Let Y = first bag is yours, and M = first bag matches.

$$P(Y) = \frac{1}{100}$$

$$\mathcal{L}(Y^c \mid M) = P(M \mid Y^c) = 0.05$$

$$\mathcal{L}(Y \mid M) = P(M \mid Y) = 1$$

$$\begin{aligned} P(Y \mid M) &= \frac{P(M \mid Y)P(Y)}{P(M)} = \frac{P(M \mid Y)P(Y)}{P(M \mid Y)P(Y) + P(M \mid Y^c)P(Y^c)} = \\ &= \frac{1 \cdot 0.01}{1 \cdot 0.01 + 0.05 \cdot 0.99} = \frac{0.01}{0.0595} \approx 0.168 \end{aligned}$$

So even though the first bag matches, it is probably not yours.

Answer: b.

$$\begin{aligned} P(Y \mid M) &= \frac{1 \cdot \frac{1}{\text{total of bags remaining}}}{1 \cdot \frac{1}{\text{total of bags remaining}} + 0.05 \cdot \frac{\text{total of bags remaining} - 1}{\text{total of bags remaining}}} \\ &\quad \text{(if 2 bags remain)} \\ &= \frac{1 \cdot \frac{1}{2}}{1 \cdot \frac{1}{2} + 0.05 \cdot \frac{1}{2}} = \frac{1}{1 + 0.05} \approx 0.9524 \end{aligned}$$

There is a 95% chance that the 99th bag is yours, given that it looks like yours.

Or, it is **19 times more likely** to be yours than not yours ($0.95/0.05=19$).